Scales and Coordinates

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Introduction

Scales and coordinates are the two layers of `ggplot2` that turn raw data into visual marks on a page. **Scales** map data values to visual properties — numbers to positions, factor levels to colours, ordered values to sizes — while **coordinates** describe the geometry of the plot area itself — Cartesian, polar, log, fixed aspect ratio. Most figure customisation happens at the scale level; coordinates change less often, but understanding the difference between them is what lets you switch between zooming, transforming, and projecting without breaking the rest of the plot.

Prerequisites

A working knowledge of `ggplot2` aesthetics and geoms, and basic familiarity with how `aes()` mappings flow through to scale choices.

Theory

Every aesthetic has a family of scales:

- `scale_x_continuous`, `scale_x_discrete`, `scale_x_log10`, `scale_x_date`, `scale_x_datetime` for x; `scale_y_*` for y.
- `scale_colour_*` and `scale_fill_*` for colour and fill, with continuous, discrete, viridis, brewer, gradient, and manual variants.
- `scale_size_*`, `scale_alpha_*`, `scale_shape_*`, `scale_linetype_*` for the remaining aesthetics.

Each scale exposes `breaks`, `labels`, `limits`, `trans`, and (for continuous) `expand` arguments that control tick placement, label formatting, axis range, monotone transformations, and padding.

Coordinates change the geometry of the panel:

- `coord_cartesian()` is the default; supplying `xlim` / `ylim` zooms without dropping data.
- `coord_flip()` swaps x and y axes after scales are applied.
- `coord_polar()` produces pie charts and radial layouts.
- `coord_fixed(ratio = 1)` enforces equal aspect ratio, essential for maps and structural diagrams.
- `coord_sf()` and `coord_map()` apply geographic projections.

The crucial distinction: `scale_x_continuous(limits = c(a, b))` *filters* observations outside `[a, b]` and recomputes statistics on the remaining data; `coord_cartesian(xlim = c(a, b))` *zooms* without filtering, leaving statistics untouched.

Assumptions

The data supplied are valid for the chosen scale (no negative values for log scales, no fractional dates for date scales).

R Implementation

```r
library(ggplot2); library(scales)
```

```r
# Log-scale y with sensible breaks
ggplot(diamonds, aes(carat, price)) +
  geom_point(alpha = 0.1) +
  scale_y_log10(labels = scales::dollar_format()) +
  theme_minimal()

# Manual colour and custom axis formatting
ggplot(mtcars, aes(wt, mpg, colour = factor(cyl))) +
  geom_point(size = 3) +
  scale_colour_manual(values = c("4" = "#2A9D8F",
                                 "6" = "#F4A261",
                                 "8" = "#6A4C93")) +
  scale_x_continuous(breaks = 1:5, labels = paste0(1:5, "K lbs")) +
  theme_minimal()

# Coord flip: horizontal bar chart
ggplot(mpg, aes(class)) +
  geom_bar(fill = "#2A9D8F") +
  coord_flip() +
  theme_minimal()

# Zoom without filtering
ggplot(diamonds, aes(carat, price)) +
  geom_point(alpha = 0.1) +
  coord_cartesian(xlim = c(0, 2)) +
  theme_minimal()
```

Output & Results

Four illustrative plots: a log-scale price axis with currency formatting, a manual colour scale with custom x-axis labels, a horizontal bar chart via `coord_flip()`, and a zoomed-in scatter that preserves the full underlying dataset for accurate statistical layers.

Interpretation

A reporting sentence: "Price is shown on a base-10 log scale (`scale_y_log10`); the linear relationship between log-price and carat reflects an approximately power-law pricing structure." Note any non-default scale or coordinate in the figure caption — a log-scale axis is easy to overlook and can mislead readers expecting linear interpretation.

Practical Tips

- Reach for `scale_y_log10()` whenever the y-axis spans orders of magnitude; constant-rate growth then reads as a straight line.
- Use `coord_cartesian()` to zoom — `scale_x_continuous(limits = ...)` *removes* the filtered data and recomputes any smoother or summary, which can mislead.
- Always label axis units; "Weight" alone is ambiguous, "Weight (1000 lbs)" is informative.
- For date axes, use `scale_x_date(date_breaks = "1 year", date_labels = "%Y")` and pick break frequencies that match the publication width.
- Reverse a scale with `scale_x_reverse()` (continuous) or `scale_x_discrete(limits = rev)` (categorical); avoid hand-flipping data frames.
- For diverging numeric scales centred at zero, combine `scale_fill_gradient2()` with explicit `low`, `mid`, `high` colours; avoid one-sided palettes that misrepresent symmetry.

R Packages Used

`ggplot2` for the scale and coordinate functions; `scales` for label formatters (`dollar_format`, `percent_format`, `comma_format`, `label_log`); `ggh4x` for nested axis labels and per-panel scale control; `cowplot` for `theme_cowplot()` paired with these scales for a publication look.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts